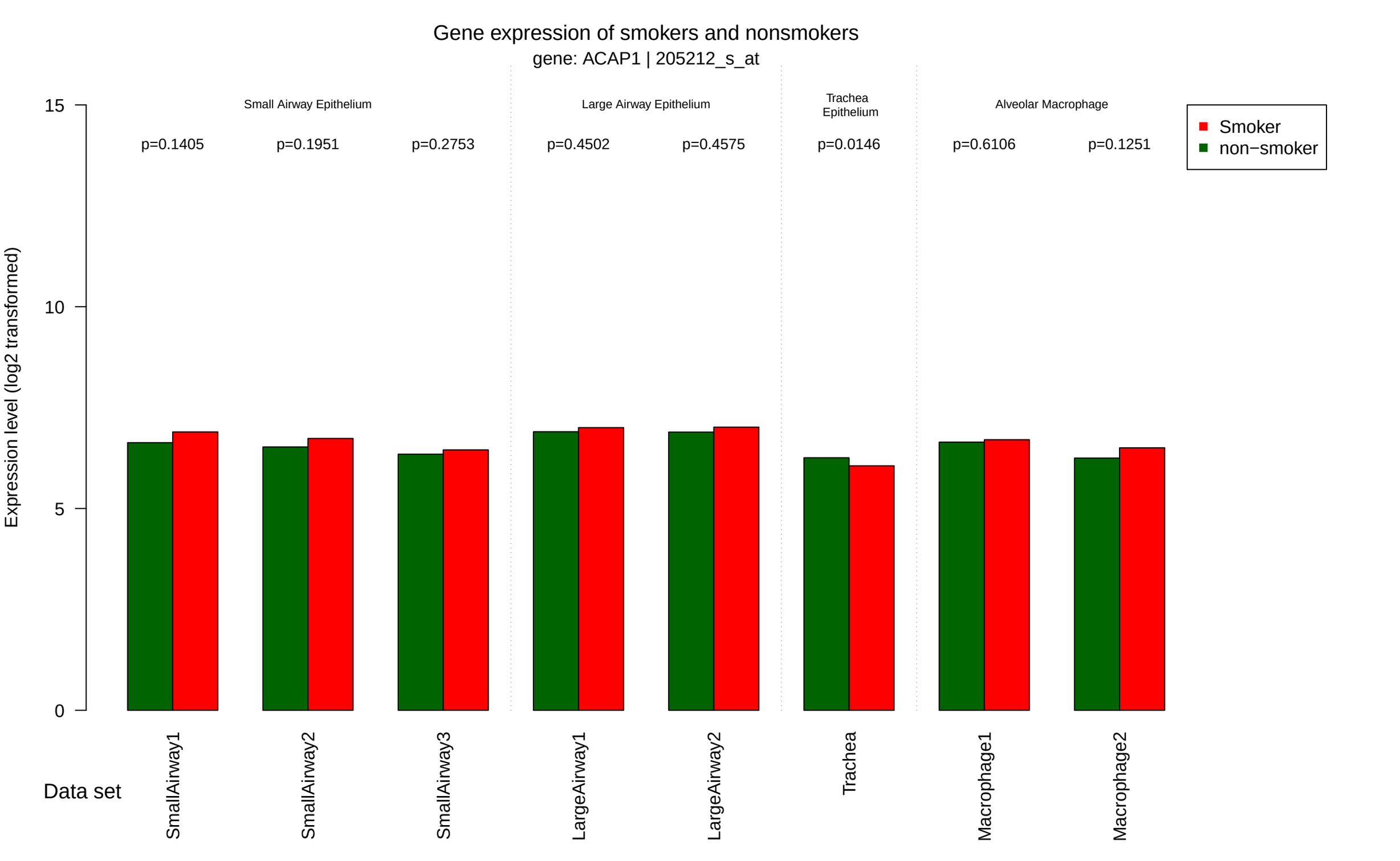
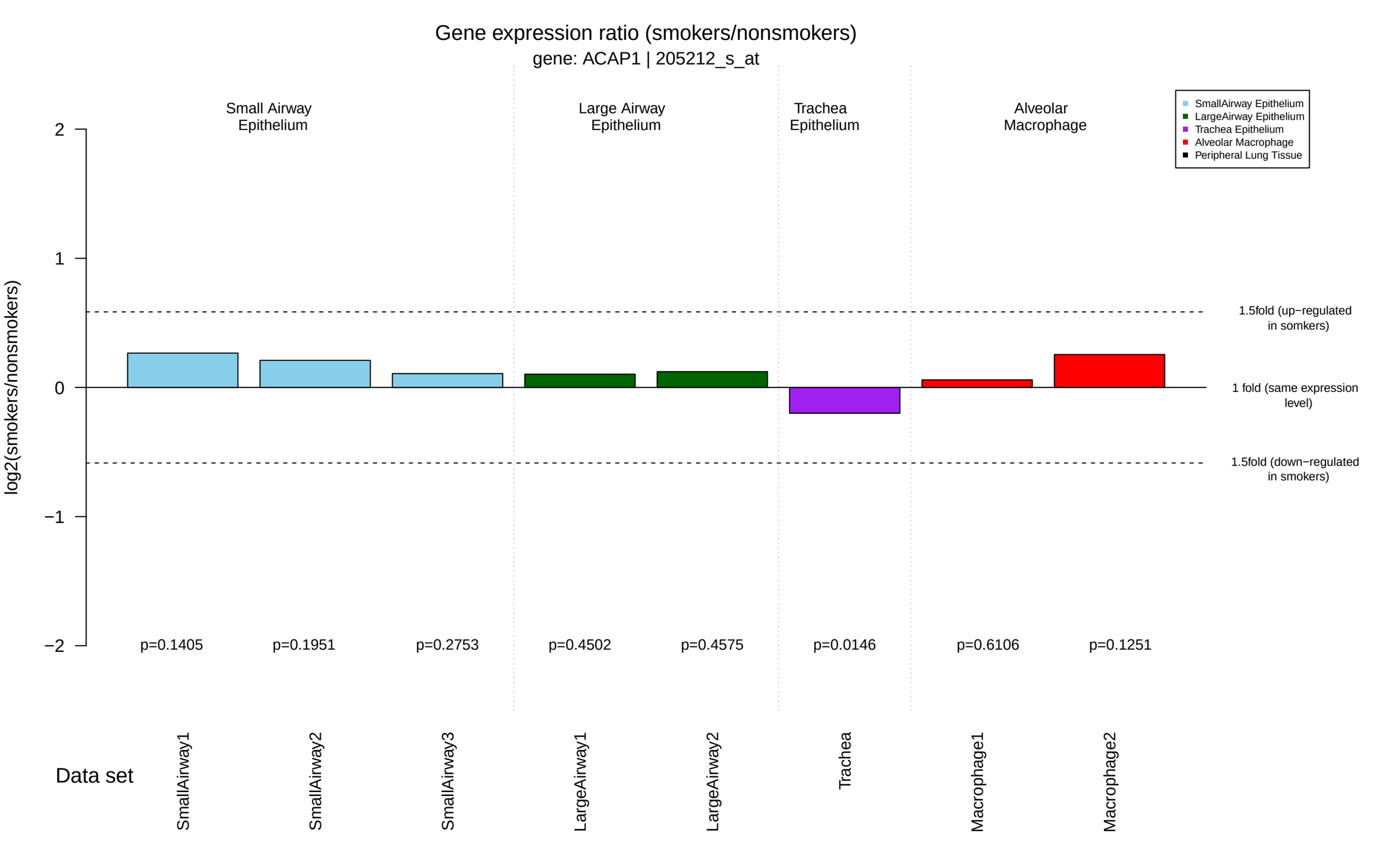
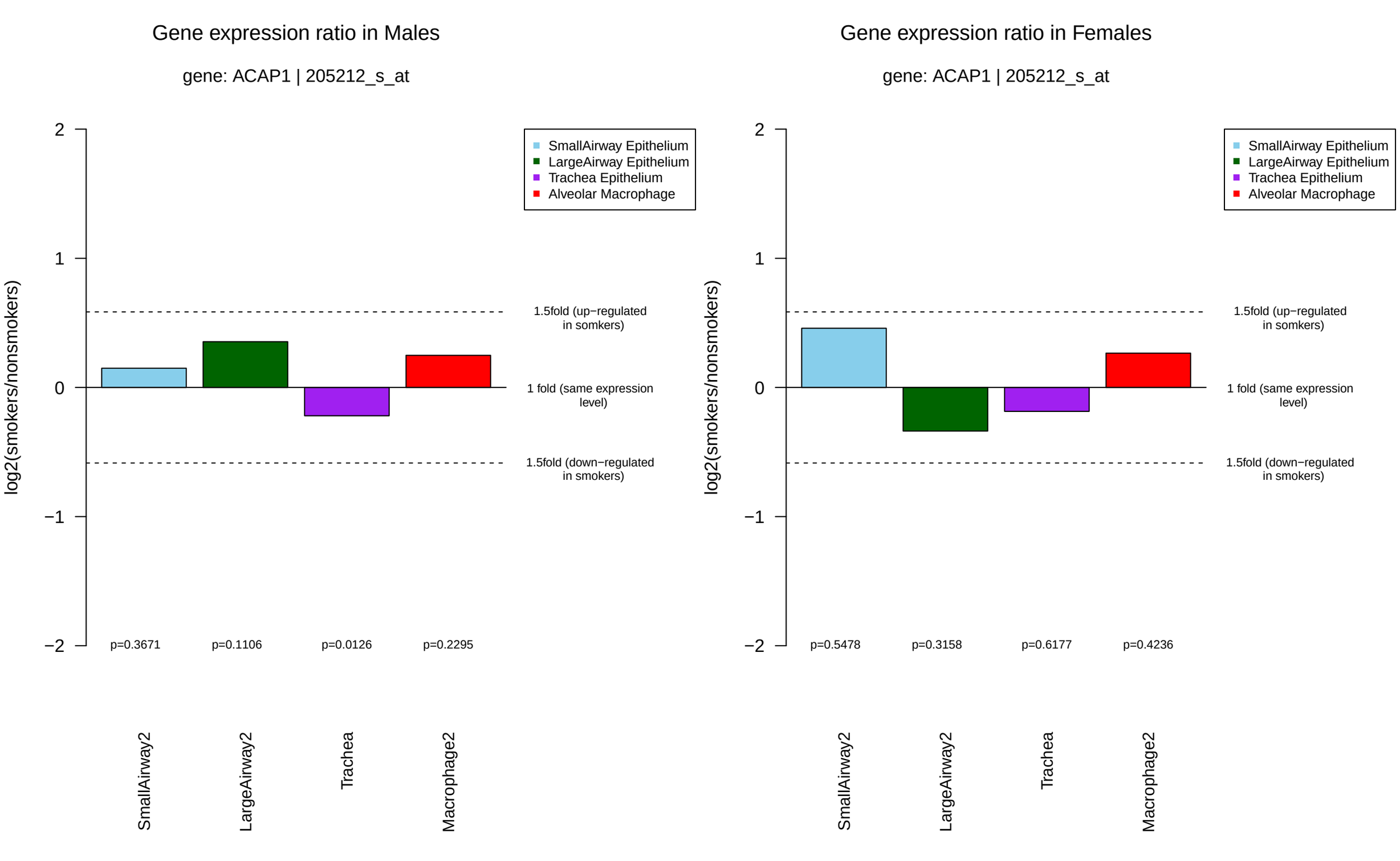
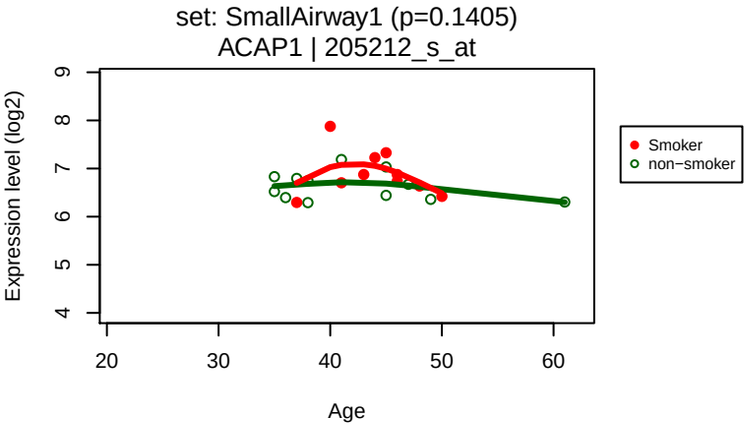
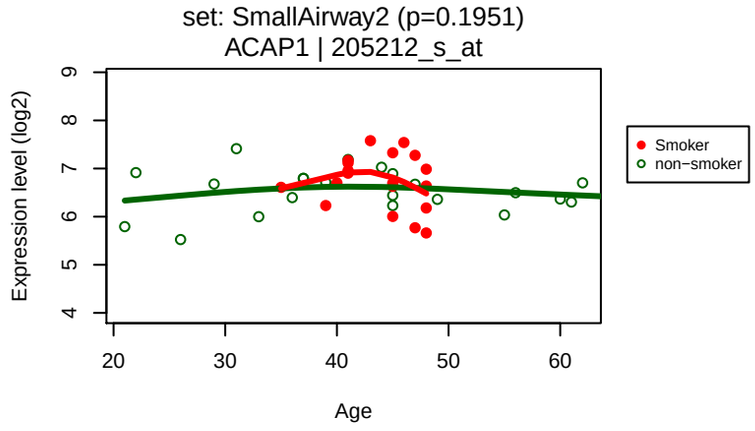
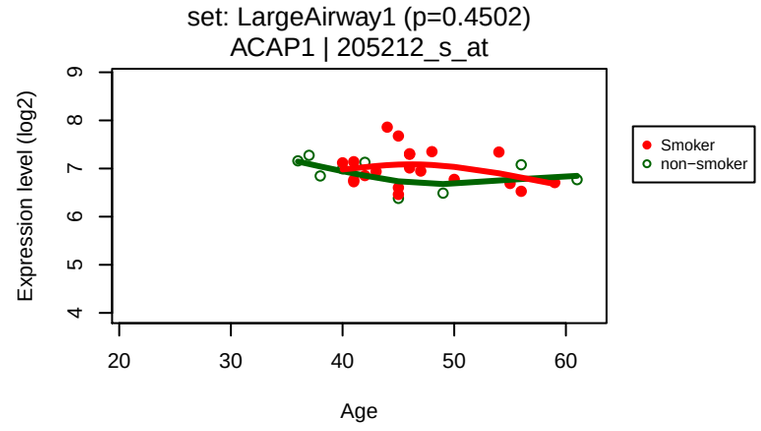

Gene expression of smokers and nonsmokers
gene: ACAP1 | 205212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1405
p=0.1951
p=0.2753
p=0.4502
p=0.4575
p=0.0146
p=0.6106
p=0.1251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

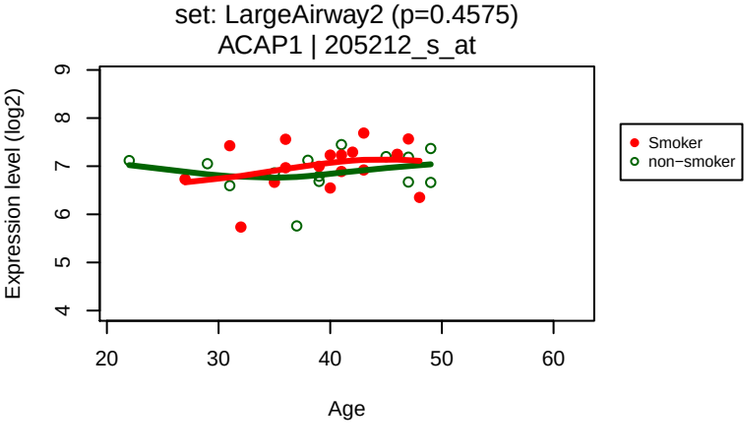

Gene expression ratio (smokers/nonsmokers)
gene: ACAP1 | 205212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1405
p=0.1951
p=0.2753
p=0.4502
p=0.4575
p=0.0146
p=0.6106
p=0.1251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

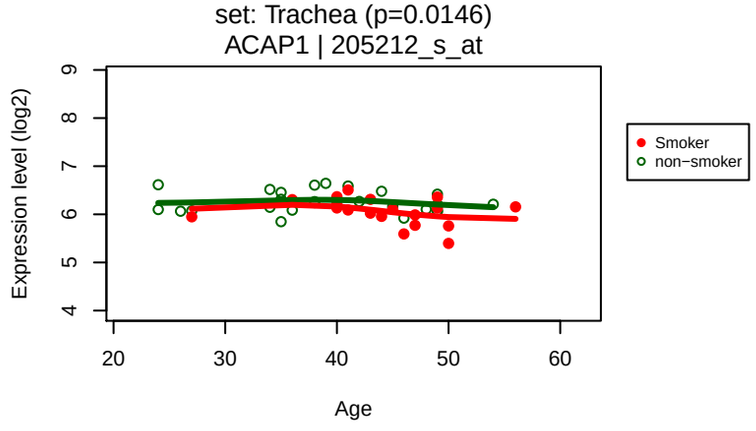

Gene expression ratio in Males
gene: ACAP1 | 205212_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3671
p=0.1106
p=0.0126
p=0.2295
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAP1 | 205212_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5478
p=0.3158
p=0.6177
p=0.4236
SmallAirway2
LargeAirway2
Trachea
Macrophage2

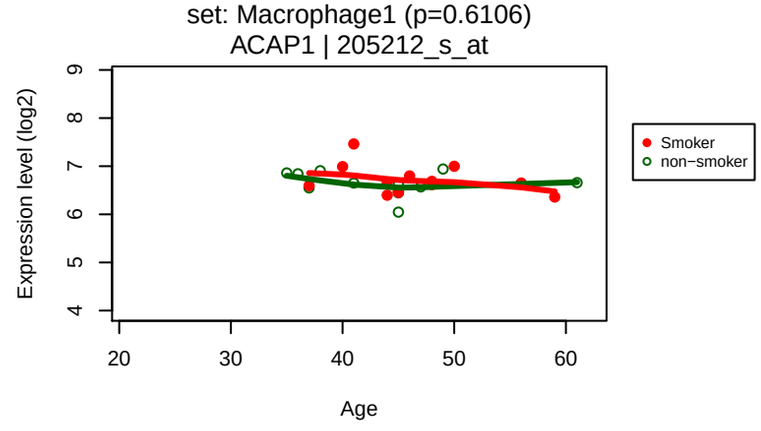

set: SmallAirway1 (p=0.1405)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker

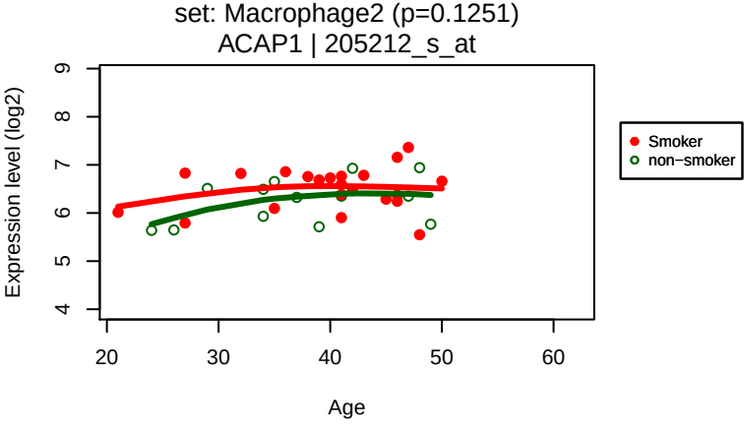

set: SmallAirway2 (p=0.1951)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker

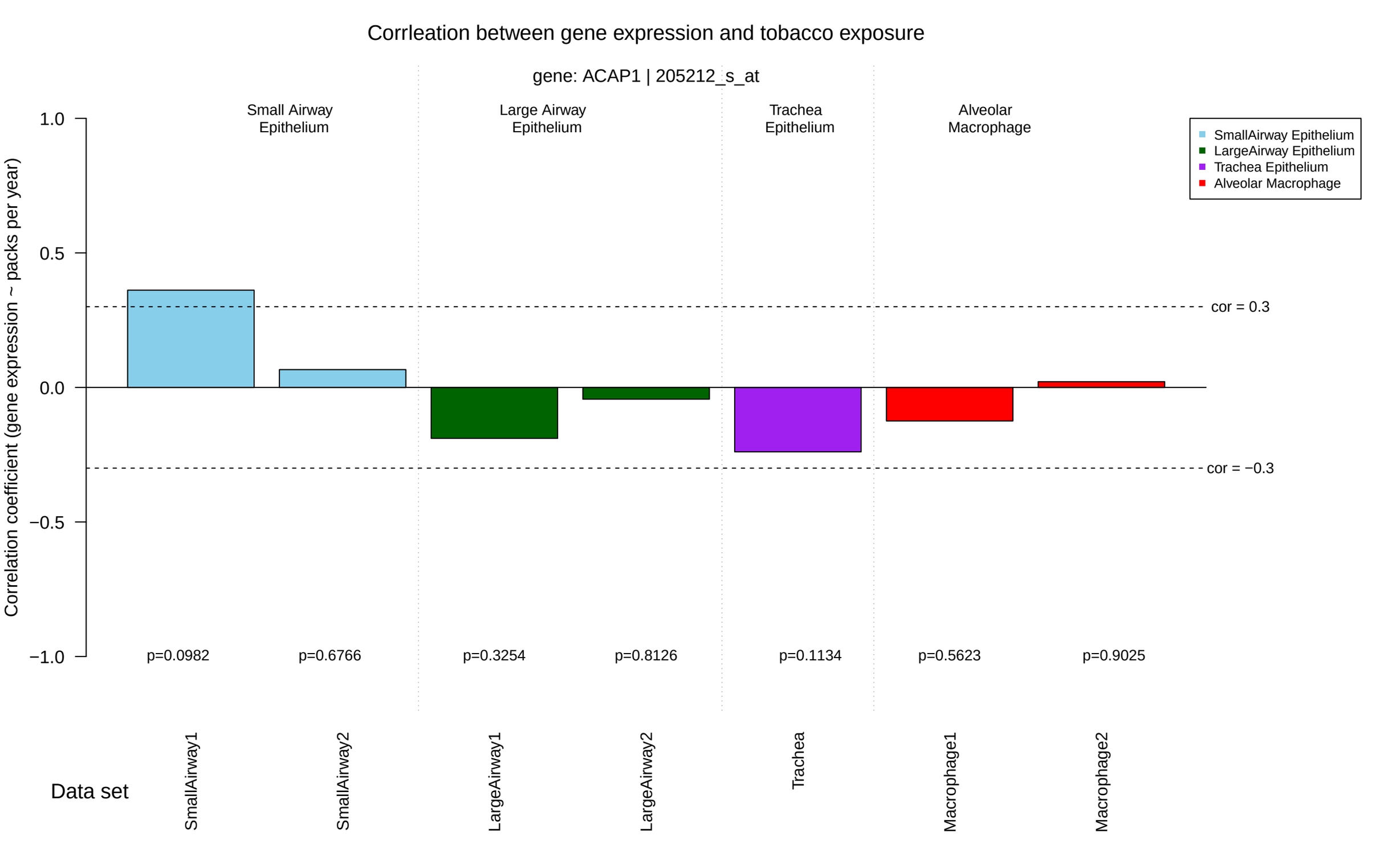

set: LargeAirway1 (p=0.4502)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4575)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0146)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6106)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1251)
ACAP1 | 205212_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAP1 | 205212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0982
p=0.6766
p=0.3254
p=0.8126
p=0.1134
p=0.5623
p=0.9025
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2